



# Development of CART model for prediction of tuberculosis treatment loss to follow up in the state of São Paulo, Brazil: A case-control study

Verena Hokino Yamaguti<sup>a,\*</sup>, Domingos Alves<sup>a,d</sup>, Rui Pedro Charters Lopes Rijo<sup>a,b,c,d</sup>,  
Newton Shydeo Brandão Miyoshi<sup>a</sup>, Antônio Ruffino-Netto<sup>a</sup>

<sup>a</sup> Ribeirão Preto Medical School, University of São Paulo, Ribeirão Preto, State of São Paulo, Brazil

<sup>b</sup> School of Technology and Management, Polytechnic Institute of Leiria, Leiria, Portugal

<sup>c</sup> Institute for Systems and Computers Engineering (INESCC), Coimbra, Portugal

<sup>d</sup> Center for Research in Health Technologies and Services (CINTESIS), University of Porto, Porto, Portugal

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## ABSTRACT

**Background:** Tuberculosis is the leading cause of infectious disease-related death, surpassing even the immunodeficiency virus. Treatment loss to follow up and irregular medication use contribute to persistent morbidity and mortality. This increases bacillus drug resistance and has a negative impact on disease control.

**Objective:** This study aims to develop a computational model that predicts the loss to follow up treatment in tuberculosis patients, thereby increasing treatment adherence and cure, reducing efforts regarding treatment relapses and decreasing disease spread.

**Methods:** This is a case-controlled study. Included in the data set were 103,846 tuberculosis cases from the state of São Paulo. They were collected using the TBWEB, an information system used as a tuberculosis treatment monitor, containing samples from 2006 to 2016. This set was later resampled into 6 segments with a 1-1 ratio. This ratio was used to avoid any bias during the model construction.

**Results:** The Classification and Regression Trees were used as the prediction model. Training and test sets accounted for 70% in the former and 30% in the latter of the tuberculosis cases. The model displayed an accuracy of 0.76, F-measure of 0.77, sensitivity of 0.80 and specificity of 0.71. The model emphasizes the relationship between several variables that had been identified in previous studies as related to patient cure or loss to follow up treatment in tuberculosis patients.

**Conclusion:** It was possible to construct a predictive model for loss to follow up treatment in tuberculosis patients using Classification and Regression Trees. Although the fact that the ideal predictive ability was not achieved, it seems reasonable to propose the use of Classification and Regression Trees models to predict likelihood of treatment follow up to support healthcare professionals in minimising the loss to follow up.

## 1. Introduction

Tuberculosis (TB) is the leading cause of infectious disease-related death, surpassing the immunodeficiency virus (HIV) [1]. TB control is based on early and appropriate diagnosis and treatment, interrupting the transmission chain [2]. The National Tuberculosis Control Plan (PNCT) is a strategy implemented by the Brazilian government. The Directly Observed Treatment Short-Course (DOTS), is a part of this strategy and has assisted in the country's reach to an 85% cure rate and has reduced the loss to follow up rate in TB treatments by 5% [2]. However, loss to follow up treatment and irregular medication use contribute to the persistence of morbidity and mortality, increasing

drug resistance cases [3,4].

TB is related to poverty and different social causes. Therefore, patients are difficult to monitor. Factors such as initial improvement, lack of patient follow-up and monitoring are all associated with poor treatment adherence [5]. After the first few months of using the correct medications during treatments, patients present improvements in their overall physical well-being and most of their symptoms are gone, which then leads to patients interrupting the treatment themselves, even though they are not cured. Social factors, treatment methodology and provided health services are factors associated with treatment loss to follow up [6–8].

In the last 10 years, computational techniques for predicting the

\* Corresponding author.

E-mail addresses: [verena.yamaguti@usp.br](mailto:verena.yamaguti@usp.br) (V. Hokino Yamaguti), [quiron@fmrp.usp.br](mailto:quiron@fmrp.usp.br) (D. Alves), [rui.rijo@ipleiria.pt](mailto:rui.rijo@ipleiria.pt) (R.P. Charters Lopes Rijo), [newton.sbm@gmail.com](mailto:newton.sbm@gmail.com) (N.S. Brandão Miyoshi), [aruffino@fmrp.usp.br](mailto:aruffino@fmrp.usp.br) (A. Ruffino-Netto).

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outcomes of TB treatment have been explored. Neural networks have been applied to predict the TB treatment loss to follow up in some cities of Espírito Santo in Brazil [9]. A logistic regression model was used to select features [10] and predict loss to follow up TB treatment [5]. Another study compared the application of different machine learning methods to classify treatment outcomes, among which the C4.5 decision tree showed better performance [11].

With a computational model that is able to identify patients that are prone to not follow-up treatment, teams would then be able to focus their efforts in reducing the rate of relapse, dissemination and loss to follow-up treatment. Although there are studies about the subject [9,10,5,11], no study has developed an effective human-readable model that can be applied by health professionals in TB treatment control. Decision trees can be represented as a set of rules that would assist healthcare professionals, the ones with and the ones without access to an information system, to control TB. Thus, this study aims to develop a CART predictive model to predict TB treatment loss to follow up cases.

## 2. Materials and methods

For the development of this case-control study, 208,620 TB treatment cases were used, all between the years of 2006 and 2016 and in the state of São Paulo. The sample had sensitive and drug-resistant tuberculosis cases. The state of São Paulo is one of 27 federal states of Brazil and is located in the southeast region of the country. It has 645 cities and an area of 248,219,491 km<sup>2</sup> [12]. It has the greatest population in the country, with around 45.9 million inhabitants [12].

The inclusion criteria considered treatment only when the outcome was either cure or loss to follow up, totaling in 180,100 cases. Data was collected through TBWEB, a system used by the public healthcare system for reporting and monitoring TB treatments in the state of São Paulo. The Health Secretary of the State of São Paulo granted access to the data and all information was previously anonymized.

Inmates and patients with invalid information for the variables in 3 were excluded. The resulting data set had 103,846 TB treatment cases, and it was split into 6 sub-samples with a 1-1 ratio of cases and controls. The ratio was chosen to reduce any class imbalance bias. The sample numbers were empirically chosen to maximize the model precision, and sampling was done following the random under-sampling technique [13]. There are 46 attributes that describe each instance in the dataset S1 File. We defined it as case when there was a treatment loss to follow up and as control when the patient was cured.

Cured patients are those with two consecutive sputum culture-negative tests within a minimum interval of 30 days after the 12th month of treatment, without clinical and radiological signs of active disease. In the case of a sputum culture-positive test in the 12th month, the treatment is prolonged for up to 24 months. The patient is then only considered cured after three consecutive negative tests with a minimum interval of 30 days and without any clinical and radiological signs of active disease. Treatment loss to follow up refers to patients who did not attend the healthcare establishments for more than 30 consecutive days after the expected date of their return, or, in the cases of supervised treatments, after the date of the last taken medication. Patients that completed the treatment but were not cured were classified as treatment failures and not addressed in this study.

The cross-validation applied to the 6 balanced folds allows us to perform a non-skewed analysis of the discrepancies between the number of cases and controls in the initial dataset. It also allows us to use the full set of available information. Since the attributes presented in the database are mostly categorical, they were transformed into indicative variables and totaled 194 binary attributes, each of these new attributes representing the presence or not of its possible categories. This transformation, called one-hot encoding, prevents any missing data from affecting the prediction process and prevents us from generating any bias due to the transformation of categories into numbers.

**Table 1**

Number of doses classifiers correlation to the treatment loss to follow up.

| Attribute                             | Loss to follow up |
|---------------------------------------|-------------------|
| Number of doses from 3rd to 6th month | −0.239080         |
| Number of doses up to the 2nd month   | −0.164101         |

On the other hand, quantitative attributes were standardized and missing values were not considered.

In the attribute analysis step, each attribute was evaluated as a predictor, either for the loss to follow-up treatment or for the cure outcome. The ranking was created using the Bagging Tree technique [14]. This method initializes and generates different decision tree models. Lastly, the importance of an attribute was given by the total entropy of the initialized trees [15] and therefore, it is possible to detect which of the attributes have the most entropy among all trees. With this, we classify all the attributes in accordance to their importance as predictors for the outcomes of TB treatment. Additionally, to avoid redundancy between some classifiers and treatment outcomes, we have inspected the correlation between them. Classifier correlation was tested using the Point-biserial correlation coefficient.

By consulting with a specialist, we were able to identify some attributes as possible tautology, such as the number of doses confirmed from the 3rd month to the 6th month and the number of doses up to the 2nd month. We analyzed the correlation of these classifiers and the treatment outcome in Table 1 to exclude any classifier that could indicate a tautology. None of them showed a correlation higher than 0.5 or lower than −0.5, and as a result, we considered these classifiers in the final model, since they could not be considered tautologies in our study dataset.

Finally, by using the most important attributes identified through the aforementioned ranking, the prediction model was created. The CART algorithm was used to develop the prediction model. **This model was selected because of its robustness against outliers and for being a decision tree. As a decision tree, the model is human-readable, and can be used either as a paper printable guideline or as an algorithm that can be implemented into a computational system,** therefore providing a model that can be used in either the absence or presence of a proper computational system infrastructure. Having it be available to those without access to an information system was taken into account because the model aims to reduce treatment loss to follow up in a country where the infrastructure for a computation system is uncertain in some locations.

The model was trained and tested with 70% and 30% of the population, from each of the segments, respectively. The model's predictive capacity evaluation was performed by using the metrics extracted for the test population. The metrics used were: Receiver Operating Characteristic (ROC) curve, Sensitivity, Specificity, Confusion Matrix and F-Measure. Decision trees have already been used by other authors as predictors of TB treatment. The current study is differentiated by the use of the CART algorithm, which is characterized by the construction of binary trees, in which each node has exactly two leaves and is more robust against outliers.

This study was submitted to Plataforma Brasil and has the number CAAE: 70947317.3.0000.5440. This study was approved by CEP (Comitê de Ética em Pesquisa, i.e., Committee of Research Ethics)/CONEP (Comissão Nacional de Ética em Pesquisa i.e., National Research Ethics Commission) of the Brazilian Government.

## 3. Results

The final population of the study contained 91,823 controls and 12,023 cases (Fig. Figure 1 ). Table Table 2 shows the distribution for the population demographic data.

Most of the patients are male adults between the ages of 20 and 39

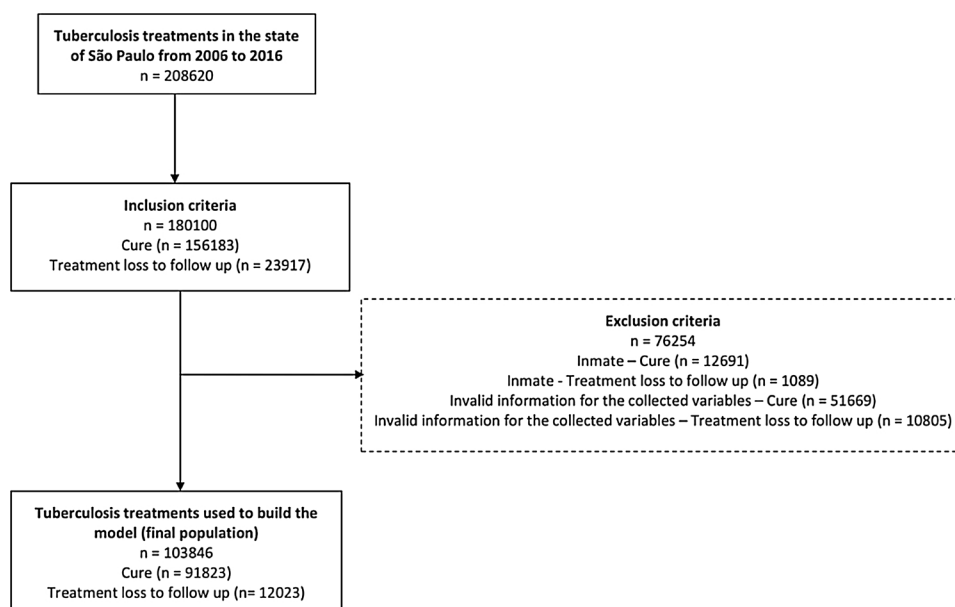


Fig. 1. Patient inclusion flowchart.

(24.67%). Less than 2% of the population represents the youth and elders.

The sampled population is mainly composed of white (39.84%) and brown (26.69%) ethnic groups. Only 0.52% of them were from an indigenous ethnic group. Most patients had basic education (less than 11 years of study) or had never studied (none).

The ranking of the 20 most important attributes that classify between cure and loss to follow up treatment can be seen in Table 3. It is possible to observe that the number of doses confirmed from the 3rd month to the 6th month is the attribute that stands out the most for identifying loss to follow up treatment cases, representing a much greater entropy when compared to other attributes. Other attributes with similar importance for classifying treatment loss to follow up are

age, total number of people residing with the patient, and the number of people who attend the exam. As follows, other variables related to social vulnerabilities and comorbidities.

Only the first 20 analyzed variables were used to generate the prediction model. The 3rd to 6th month doses were excluded as well as the doses from the 1st and 2nd month. The number of variables used were determined empirically, in order to maximize the defined quantitative metrics. Each of the segments that were used for training contained 12,023 loss to follow-up treatment cases and 12,023 cure cases. To train the model, 16,842 (70%) cases from each of the 6 segments of the population were used. Subsequently, the model test used 7218 (30%) individuals from each segment.

The generated model is visually represented by a tree, which can be

Table 2

Distribution of patients treated by age group, sex, ethnic groups and scholary.

| Variables | Value          | Control (N = 91, 823) | Case (N = 12, 023) | Total (N = 103, 846) |
|-----------|----------------|-----------------------|--------------------|----------------------|
| Age group | < 1 year       | 385 (0.37%)           | 33 (0.03%)         | 418 (0.40%)          |
|           | 1–4 years      | 1085 (1.04%)          | 51 (0.05%)         | 1136 (1.09%)         |
|           | 5–9 years      | 882 (0.85%)           | 29 (0.03%)         | 911 (0.88%)          |
|           | 10–14 years    | 1550 (1.49%)          | 49 (0.05%)         | 1599 (1.54%)         |
|           | 15–19 years    | 6109 (5.88%)          | 733 (0.71%)        | 6842 (6.59%)         |
|           | 20–29 years    | 21,735 (20.93%)       | 3882 (3.74%)       | 25,617 (24.67%)      |
|           | 30–39 years    | 19,115 (18.41%)       | 3451 (3.32%)       | 22,566 (21.73%)      |
|           | 40–49 years    | 17,086 (16.45%)       | 2253 (2.17%)       | 19,339 (18.62%)      |
|           | 50–59 years    | 13,440 (12.94%)       | 1009 (0.97%)       | 14,449 (13.91%)      |
|           | 60–69 years    | 6691 (6.44%)          | 344 (0.33%)        | 7035 (6.77%)         |
|           | 70–79 years    | 2863 (2.76%)          | 143 (0.14%)        | 3006 (2.89%)         |
|           | > = 80 years   | 882 (0.85%)           | 46 (0.04%)         | 928 (0.89%)          |
| Sex       | Female         | 32,107 (30.92%)       | 2984 (2.87%)       | 35,091 (33.79%)      |
|           | Male           | 59,716 (57.50%)       | 9039 (8.70%)       | 68,755 (66.21%)      |
| Ethnic    | Yellow         | 880 (0.85%)           | 74 (0.07%)         | 954 (0.92%)          |
|           | White          | 41,372 (39.84%)       | 4445 (4.28%)       | 45,817 (44.12%)      |
|           | Indigenous     | 541 (0.52%)           | 77 (0.07%)         | 618 (0.60%)          |
|           | Brown          | 27,715 (26.69%)       | 4154 (4.00%)       | 31,869 (30.69%)      |
|           | Black          | 9023 (8.69%)          | 1530 (1.47%)       | 10,553 (10.16%)      |
|           | No information | 12,292 (11.84%)       | 1743 (1.68%)       | 14,035 (13.52%)      |
| Scholary  | None           | 3681 (3.54%)          | 301 (0.29%)        | 3982 (3.83%)         |
|           | 1–3 years      | 8355 (8.05%)          | 1010 (0.97%)       | 9365 (9.02%)         |
|           | 4–7 years      | 25,055 (24.13%)       | 3838 (3.70%)       | 28,893 (27.82%)      |
|           | 8–11 years     | 28,437 (27.38%)       | 3488 (3.36%)       | 31,925 (30.74%)      |
|           | 12–14 years    | 6037 (5.81%)          | 429 (0.41%)        | 6466 (6.23%)         |
|           | > = 15 years   | 2887 (2.78%)          | 168 (0.16%)        | 3055 (2.94%)         |
|           | No information | 17,371 (16.73%)       | 2789 (2.69%)       | 20,160 (19.41%)      |

**Table 3**  
Ranking of the most important attributes.

| Ranking | Sum of entropy of trees | Attribute                                                               |
|---------|-------------------------|-------------------------------------------------------------------------|
| 1°      | 0.077879                | Number of doses from 3rd to 6th month                                   |
| 2°      | 0.039481                | Age in completed years                                                  |
| 3°      | 0.035257                | Number of doses up to the 2nd month                                     |
| 4°      | 0.032344                | Total number of people residing with the patient                        |
| 5°      | 0.031436                | Number of people residing with the patient who attended the examination |
| 6°      | 0.024282                | Non-drug addicts                                                        |
| 7°      | 0.020861                | Negative HIV test                                                       |
| 8°      | 0.017036                | Drug addicts                                                            |
| 9°      | 0.016322                | Total 8–11 years of study completed                                     |
| 10°     | 0.016327                | Chest XR as suspected TB                                                |
| 11°     | 0.014258                | Type of occupation as unemployed                                        |
| 12°     | 0.015270                | Discovery of the case was in outpatient demand                          |
| 13°     | 0.016537                | Total 4–7 years of study completed                                      |
| 14°     | 0.014184                | Self-reported race or color as brown                                    |
| 15°     | 0.013968                | Type of treatment as supervised                                         |
| 16°     | 0.014153                | Type of occupation as other                                             |
| 17°     | 0.014522                | Sputum culture not performed                                            |
| 18°     | 0.013160                | Necropsy examination not performed                                      |
| 19°     | 0.013338                | Discovery of the case in Urgency and Emergency Unit                     |
| 20°     | 0.012944                | Race or self color reported by the patient as white                     |

seen in Fig. Figure 2 of this article. Table Table 4 features the averages for accuracy, *F*-Measure, sensitivity and specificity metrics for the test folders that were used to quantitatively evaluate the developed predictive model. From which it is possible to highlight the high sensitivity and *F*-Measure presented by the model.

Table Table 5 shows the total count number of false positives, false negatives, true positives and true negatives obtained from the test folders. In it we highlight the true positive and true negative numbers.

Fig. [Figure 3](#) shows the obtained ROC curve for each of the test folds of the model contrasted by the dashed line that represents a model that randomly classifies the cases. We can observe that the curve and its area under it is similar for all the analyzed segments, the average Area Under the Curve (AUC) can be viewed between segments 0.73.

## 4. Discussion

The decision tree created can be used as a guideline among health professionals to detect treatment loss to follow up. The accuracy of the model is similar to other models already presented in the literature [11]. For health professionals, the decision tree is a much more intuitive and understandable approach than a statistical model, since it can be visualized and explored by anyone, as well as be applied without a computational system. To exemplify, the decision tree for this study can be visualized in Fig. [Figure 2](#).

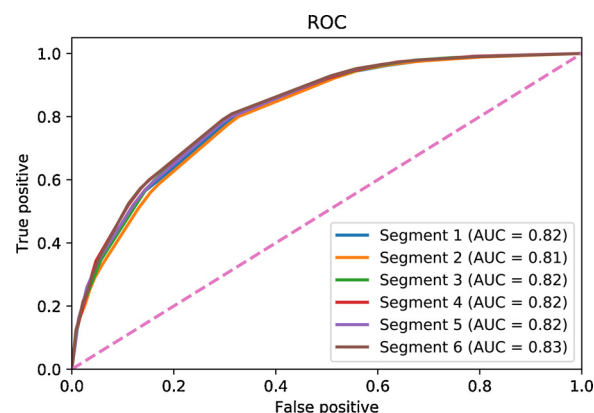
The model emphasizes the relationship between several variables already identified in the literature as patient loss to follow-up treatment

**Table 4**  
Quantitative metrics used to evaluate the predictive model.

| Metrics           | Value  |
|-------------------|--------|
| Accuracy          | 0.7601 |
| <i>F</i> -measure | 0.7703 |
| Sensitivity       | 0.8048 |
| Specificity       | 0.7156 |

**Table 5**  
Matrix of confusion for the predictive model.

|       | Positive | Negative |
|-------|----------|----------|
| True  | 17,413   | 15,509   |
| False | 6163     | 4223     |

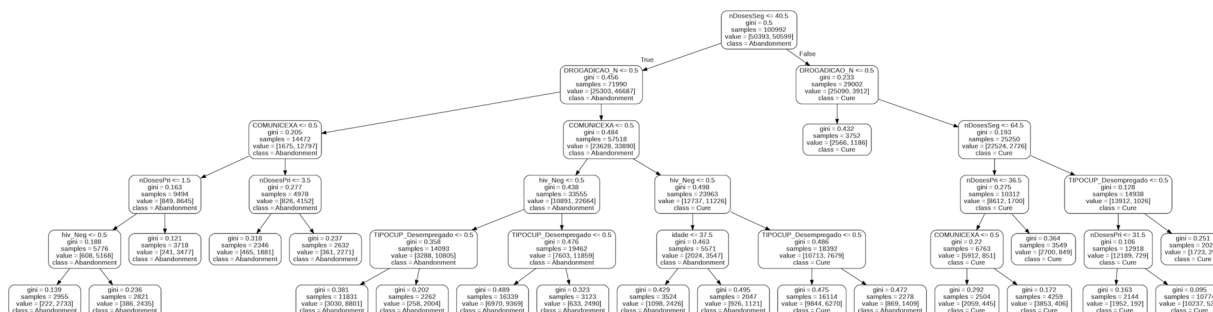


**Fig. 3.** Folder ROC curve.

or patient cure. Among which are patient ethnicity, unemployment, schooling, drug addiction and age [10,5,11]. However, other factors not raised by the authors appear as factors of great importance to predict the treatment loss to follow up that are cited in this study.

Among these variables, we have highlighted the variables that represent the number of doses received by the patient up until the 2nd month as well as up until the 3rd to the 6th month. Although some specialists believe that it is possible that these variables represent a tautology, they do not have a high enough correlation to be considered one. This further emphasizes that what determines a treatment's success is the frequency of doses and correct medication taken throughout it and not the number of doses taken throughout the treatment. The monitoring of dose numbers taken and its frequency can act as an alarm trigger, notifying health professionals automatically.

In general, our model presented a lower number of false positives than other models presented in the literature [5]. This is mainly due to the balance between cure and treatment loss to follow up cases. This



**Fig. 2.** Decision tree.

approach reduces the bias generated by the imbalance classes. On the other hand, our model failed to adequately optimize the accuracy indexes already presented by similar models in the literature [11]. Thus, we believe experimentation with other models may be appropriate.

The final model has an accuracy higher than the logistic regression model and the C4.5 presented in the literature [5,11]. We also showed a better value for the *F*-measure than the C4.5 model [11]. However, the C4.5 model [11] presents a better result for the ROC curve.

## 5. Conclusion

It was possible to build the predictive model of treatment loss to follow up of tuberculosis through CART. During model development, new features for TB treatment loss to follow up predictors were identified. Although the fact that the ideal predictive ability was not achieved, it seems reasonable to propose the use of CART models. We aim in future studies to improve the model accuracy and implement the model in an informational system for tuberculosis treatment and management.

## Summary Points

What was already known on the topic:

- Treatment loss to follow up and irregular medication use contribute to increased bacillus drug resistance and has a negative impact on disease control.
- Other studies apply different models to predict the TB treatment outcome.
- There is evidence that, among different models, decision trees present better performance.

What this study added to our knowledge:

- The CART model was able to perform equally or better than other models in the literature, additionally providing a human-readable model that can be used without an informational system.
- Features such as age, the number of doses taken by the patient between the 3rd and 6th month, the number of people resident with the patient were ranked among the feature with greater entropy to predict treatment loss to follow up.
- The number of doses taken by the patient and the treatment outcome presented a correlation lower than expected, discarding the possibility of a possible tautology, as suggested by specialists.

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## Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ijmedinf.2020.104198>.

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